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Thermodynamic assessments of the Cu–B and Cu–Tm systems

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ABSTRACT

By using the calculation of phase diagrams (CALPHAD) method, the thermodynamic assessments of the Cu–B and Cu–Tm systems were carried out based on the available experimental data including thermodynamic properties and phase equilibria. Gibbs free energies of the liquid, fcc and β rhombohedral phases in the Cu–B system and the liquid, fcc and hcp phases in the Cu–Tm system were modeled by the subregular solution model with the Redlich–Kister formula, and those of the intermetallic compounds (Cu_5Tm, Cu_9Tm_2, Cu_7Tm_2, Cu_2Tm and CuTm compound) in the Cu–Tm system were described by the sublattice model. An agreement between the present calculated results and experimental data was obtained.

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1. Introduction

Copper base alloys have been widely used in many application fields because of their good combination of high thermal and electrical conductivities and high strength. In particular, high performance Cu base alloys are required in the field of electronic materials, such as substrate and lead frame in the printed board, interconnection and so on, because the electronic packaging has a tendency to miniaturization [1,2]. Alloying of rare earth elements into copper combined with the proper heat treatment may help to achieve both high strength and high electrical conductivity. Moreover, the intermetallic compounds formed between transition metals and rare earth elements are of particular importance for high performance functional materials, such as permanent magnet and hydrogen storage materials [3,4].

The CALPHAD method, which is a powerful tool to cut down on cost and time during development of materials [5], effectively provides a clear guideline for materials design. In recent years, the present authors have developed a thermodynamic database of Cu-based alloys within the framework of CALPHAD method [6–11], which is important for the design of Cu-based alloys. As part of development of thermodynamic database of the Cu-based alloys, this work is to present the thermodynamic assessments of the Cu–B and Cu–Tm binary systems based on the available experimental data by means of the CALPHAD method.

2. Thermodynamic model

The stable solid phases and the models used in the Cu–B and Cu–Tm systems [12] are listed in Table 1.

2.1. Solution phases

Gibbs free energies of the liquid, fcc and β rhombohedral phases in the Cu–B system and the liquid, fcc and hcp phases in the Cu–Tm system were described by the subregular solution model with the Redlich–Kister formula [13]. The molar Gibbs free energy of each solution phase in the Cu–B system is given as follows:

$$G_m^{\phi} = \sum_{i=\mathrm{Cu,B}} {}^0G_i^{\phi} x_i + RT \sum_{i=\mathrm{Cu,B}} x_i \ln x_i + x_{\mathrm{Cu}} x_{\mathrm{B}} \sum_{m=0}^{n} {}^mL_{\mathrm{Cu,B}}^{\phi} (x_{\mathrm{B}} - x_{\mathrm{Cu}})^m, \quad (1)$$

Similarly, the molar Gibbs free energy of each solution phase in the Cu–Tm system is given as follows:

$$G_m^{\phi} = \sum_{i=\mathrm{Cu,Tm}} {}^0G_i^{\phi} x_i + RT \sum_{i=\mathrm{Cu,Tm}} x_i \ln x_i + x_{\mathrm{Cu}} x_{\mathrm{Tm}} \sum_{m=0}^{n} {}^mL_{\mathrm{Cu,Tm}}^{\phi} (x_{\mathrm{Cu}} - x_{\mathrm{Tm}})^m, \quad (2)$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of pure component i taken from SGTE database [14]; x_i represents the mole fraction of the component i; R is the gas constant and T the absolute temperature; ${}^mL_{\mathrm{Cu,B}}^{\phi}$ and ${}^mL_{\mathrm{Cu,Tm}}^{\phi}$ are the binary interaction parameters and are,

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Table 1
The stable solid phases and the models used in the Cu–B and Cu–Tm systems.

System	Phase	Prototype	Strukturbericht designation	Modeling phase	Used models
Cu–B	(Cu)	Cu	A1	Fcc	SSM
	(β B)	β B		β Rhombohedral	SSM
Cu–Tm	(Cu)	Cu	A1	Fcc	SSM
	Cu_5Tm	$AuBe_5$	$C15_b$	$(Cu)_5(Tm)$	SM
	Cu_9Tm_2			$(Cu)_9(Tm)_2$	SM
	Cu_7Tm_2			$(Cu)_7(Tm)_2$	SM
	Cu_2Tm	$CeCu_2$		$(Cu)_2(Tm)$	SM
	CuTm	CsCl	B2	(Cu) (Tm)	SM
	(Tm)	Mg	A3	Hcp	SSM

Note: SSM, subregular solution model and SM, sublattice model.

respectively, expressed as:

$$^{m}L^{\phi}_{\mathrm{Cu,B}} = a + bT, \quad (3)$$

and $$^{m}L^{\phi}_{\mathrm{Cu,Tm}} = a' + b'T, \quad (4)$$

where the values of a, b and a', b' are evaluated in the present work based on the experimental data.

2.2. Intermetallic compounds

The intermetallic compounds (Cu_5Tm, Cu_9Tm_2, Cu_7Tm_2, Cu_2Tm and CuTm) in the Cu–Tm system are treated as stoichiometric compounds. The Gibbs free energy of the phase Cu_pTm_q is formulated as:

$$\Delta G_{\mathrm{f}}^{\mathrm{Cu}_p\mathrm{Tm}_q} = G_m^{\mathrm{Cu}_p\mathrm{Tm}_q} - p^0 G_{\mathrm{Cu}}^{\mathrm{fcc}} - q^0 G_{\mathrm{Tm}}^{\mathrm{hcp}}, \quad (5)$$

where $\Delta G_{\mathrm{f}}^{\mathrm{Cu}_p\mathrm{Tm}_q}$ represents the Gibbs free energy of formation per mole of formula unit Cu_pTm_q, which is described as

$$\Delta G_{\mathrm{f}}^{\mathrm{Cu}_p\mathrm{Tm}_q} = a'' + b''T + c''T\ln T, \quad (6)$$

where the values of a'', b'' and c'' are evaluated in the present work.

3. Experimental information

3.1. Cu–B system

The Cu–B phase diagram with the simple eutectic type and no intermediate phases was investigated by various researchers [15–21]. Rexer and Petzow [18] reported the composition (13.3 at.% B) and the temperature (1013 °C) of the eutectic reaction L ↔ (Cu) + (B). According to Smiryagin and Kvurt's work [16], the solid solubility of B in (Cu) is ~0.29 at.% B at the eutectic temperature. Rexer and Petzow [18] reported the maximum solubility of Cu in (B) at 950 °C to be 2.8 at.% Cu. Moreover, Wald and Stormont [17] and Rexer and Petzow [18] studied the liquidus by means of thermal analysis. However, the liquidus at high temperatures has not been well established. The phase diagram of the Cu–B system reviewed by Chakrabarti and Laughlin [22] is shown in Fig. 1.

Jacob et al. [23] measured the activity of B in the liquid Cu–B alloy at 1450 °C with reference states of Cu (liquid) and B (β rhombohedral), and then derived the activity of B with respect to Cu (liquid) and B (liquid), and the activity of Cu with respect to Cu (liquid) and B (β rhombohedral) at the same temperature, respectively. In addition, Yukinobu et al. [24] determined the activity of B in the binary Cu–B melt at low concentrations by electromotive force measurement. However, the values reported by Yukinobu et al. [24] are significantly higher than those obtained by Jacob et al. [23], possibly due to presence of some electronic conductivity in the electrolyte [23]. Therefore, we used the data by Jacob et al. [23] in the optimization, the data by Yukinobu et al. [24] were only used in the comparison.

3.2. Cu–Tm system

Bochvar and Lysova [25] studied the solubility of Tm in the (Cu) fcc phase by metallography, electrical resistance and hardness measurements, and reported an eutectic reaction of the L ↔ (Cu) + Cu_5Tm at 850–870 °C with the maximum solubility of 0.02–0.03 at.% Tm at the eutectic temperature. The studies of the structures of the intermediate phases were carried out by various investigators, as follows: Cu_5Tm is isostructural to $AuBe_5$ [26], Cu_2Tm crystallizes with the orthorhombic $CeCu_2$-type structure [27] and CuTm with the cubic CsCl structure [28,29]. However, the existence of the Cu_9Tm_2 and Cu_7Tm_2 phases is proposed solely on the basis of the presence of similar phases in the Cu–Gd [30], Cu–Dy [31] and Cu–Er [32] systems. Moreover, melting and eutectic temperatures in the Cu–lanthanide systems, in general, are known to vary systematically across the lanthanide series. Accordingly, the invariant temperatures of above-mentioned Cu–Tm intermediate phases were estimated by extrapolation of corresponding data for those Cu–lanthanide systems for which experimental phase diagrams are already known, using the methods described by Gschneidner and Calderwood [33]. The Cu–Tm equilibrium diagram was then estimated from the extrapolated invariant temperatures. The phase diagram of the Cu–Tm system reviewed by Subramanian and Laughlin [34] is shown in Fig. 2.

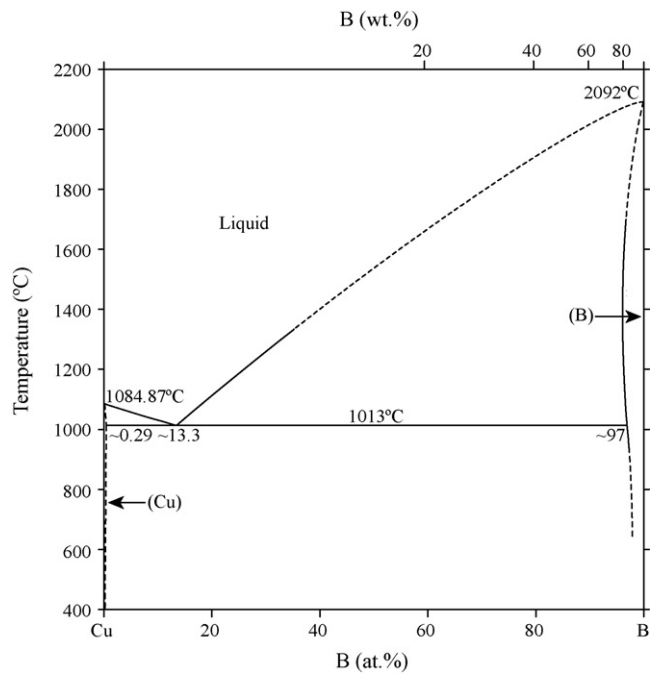

Fig. 1. The Cu–B phase diagram reviewed by Chakrabarti and Laughlin [22].

Recently, Meschel and Kleppa [35] measured the standard enthalpies of formation of the Cu_5Tm and Cu_2Tm phases by high temperature direct synthesis calorimetry at 1100 ± 2 °C. Moreover, the measurement of the enthalpy for liquid phase at 1250 °C was carried out by Nikolaenko [36] using a self-made heat flux high temperature isoperibolic calorimeter.

4. Results and discussion

The optimization was carried out using the PARROT module in the Thermo-Calc program developed by Sundman et al. [37]. The phase diagram and thermochemical data were used as input to the program. Each piece of selected information was given a certain weight by personal judgment, and verified by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits. All the evaluated parameters of each phase in the Cu–B and Cu–Tm systems are listed in Table 2.

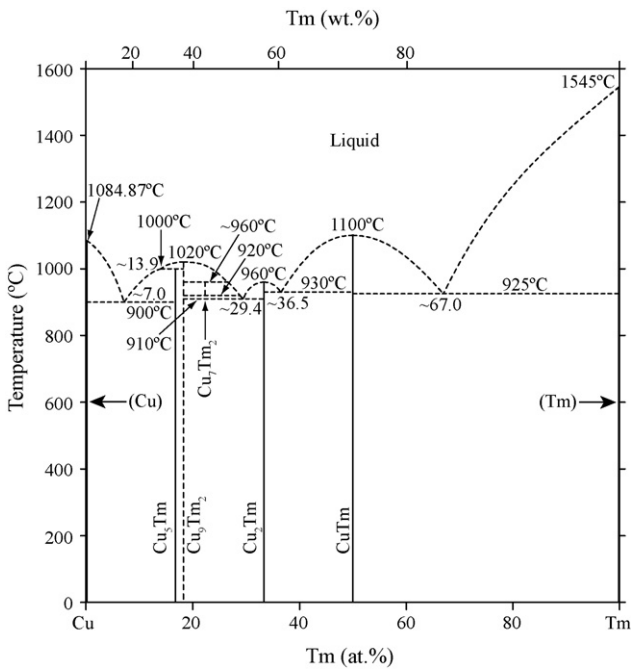

Fig. 2. The Cu–Tm phase diagram reviewed by Subramanian and Laughlin [34].

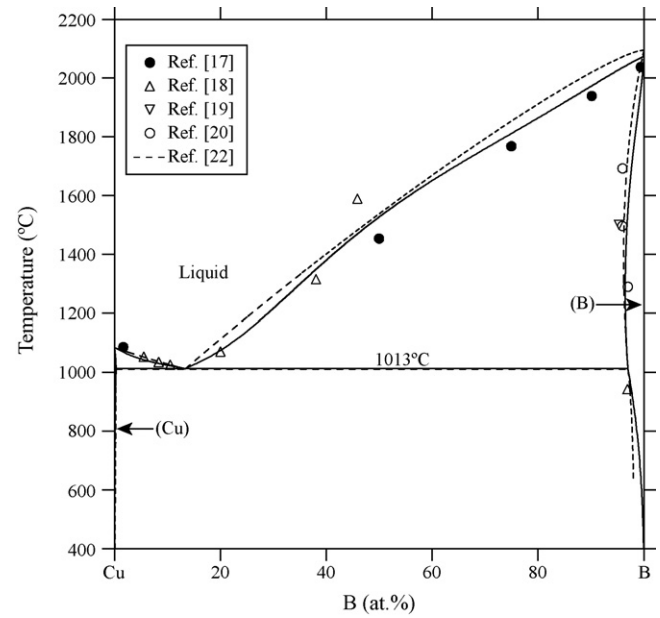

Fig. 3. The calculated Cu–B phase diagram with experimental data [17–20] and the phase diagram reviewed by Chakrabarti and Laughlin [22].

Table 2
Thermodynamic parameters in the Cu–B and Cu–Tm systems.

System	Parameters in each phase (J/mol)
Cu–B	Liquid phase, formula (B, Cu): ${}^{0}L^{liq}_{B,Cu} = -3156.2$ ${}^{1}L^{liq}_{B,Cu} = 14390.26 - 9.98T$ ${}^{2}L^{liq}_{B,Cu} = 42620 - 20T$ Fcc phase, formula (B, Cu): ${}^{0}L^{fcc}_{B,Cu} = 29600$ β Rhombohedral phase, formula (B, Cu): $G^{\beta B}_{Cu} = G^{fcc}_{Cu} + 5000$ ${}^{0}L^{\beta B}_{B,Cu} = 34386$
Cu–Tm	Liquid phase, formula (Cu, Tm): ${}^{0}L^{liq}_{Cu,Tm} = -69000 + 9T$ ${}^{1}L^{liq}_{Cu,Tm} = -28520 + 0.6T$ ${}^{2}L^{liq}_{Cu,Tm} = 13300 - 9.26T$ Fcc phase, formula (Cu, Tm): $G^{fcc}_{Tm} = G^{hcp}_{Tm} + 5000$ ${}^{0}L^{fcc}_{Cu,Tm} = 25000$ Hcp phase, formula (Cu, Tm): ${}^{0}L^{hcp}_{Cu,Tm} = 25000$ Cu_5Tm phase, formula $(Cu)_5(Tm)_1$: $\Delta G_f^{Cu_5Tm} = -19090 + 39.866T - 5T\ln T$ Cu_9Tm_2 phase, formula $(Cu)_9(Tm)_2$: $\Delta G_f^{Cu_9Tm_2} = -22760 + 70.6T - 9T\ln T$ Cu_7Tm_2 phase, formula $(Cu)_7(Tm)_2$: $\Delta G_f^{Cu_7Tm_2} = -6685.2 - 7.6T$ Cu_2Tm phase, formula $(Cu)_2(Tm)_1$: $\Delta G_f^{Cu_2Tm} = -25600 + 86T - 11.326T\ln T$ CuTm phase, formula $(Cu)_1(Tm)_1$: $\Delta G_f^{CuTm} = -16260 - 2.9T$

4.1. Cu–B system

Fig. 3 shows the calculated Cu–B phase diagram compared with selected experimental data [17–20] and the phase diagram reviewed by Chakrabarti and Laughlin [22]. The calculated phase diagram is in good agreement with the experimental results. The invariant reaction in the Cu–B system is listed in Table 3.

As shown in Fig. 4, the calculated activity of B in the liquid Cu–B alloy at 1450 °C is in agreement with the experimental data [23]. The calculated activity of B at 1200 °C and the experimental data by Yukinobu et al. [24] are also presented in Fig. 4 just for comparison. Figs. 5 and 6 show the calculated activity of B and Cu at 1450 °C compared with the data derived by Jacob et al. [23], respec-

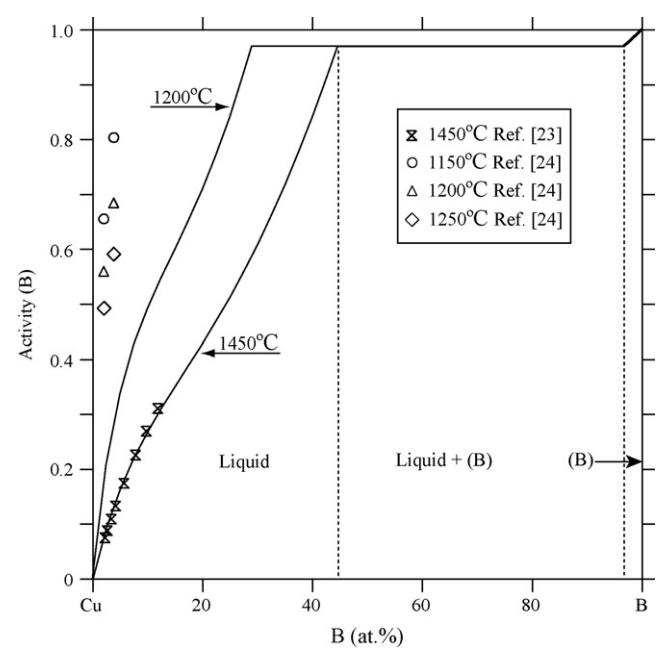

Fig. 4. The calculated activity of B at 1450 °C and 1200 °C with respect to Cu (liquid) and B (β rhombohedral) compared with experimental data [23,24].

Table 3
Invariant reaction in the Cu–B system.

Reaction	Reaction type	Composition (at.% B)			T (°C)	Ref.
L ↔ (Cu) + (B)	Eutectic	10.7			1060 ± 2	[15]
		10.7	~0.29		1021 ± 2	[16]
		≥10.7		~97	1008 ± 4	[17]
		~13.3			1013	[18]
		18-19			1010 ± 1	[20]
		13.3	0.29	97.0	1013	This work

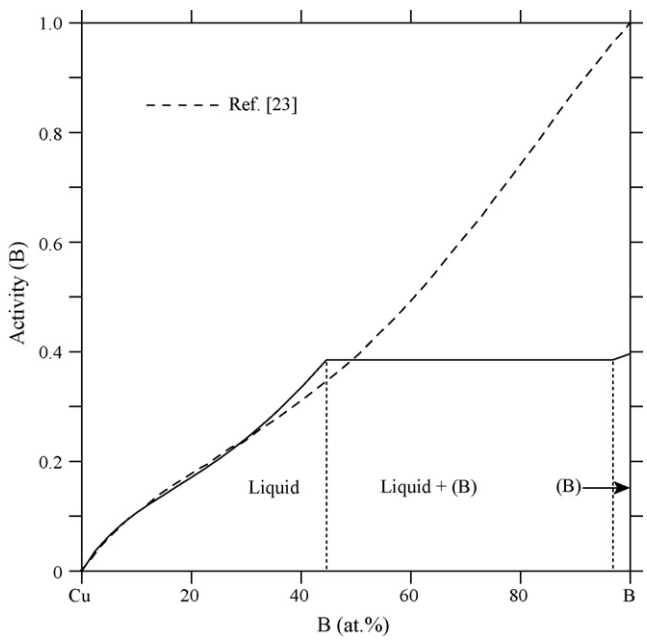

Fig. 5. The calculated activity of B at 1450 °C with respect to Cu (liquid) and B (liquid) compared with derived data by Jacob et al. [23].

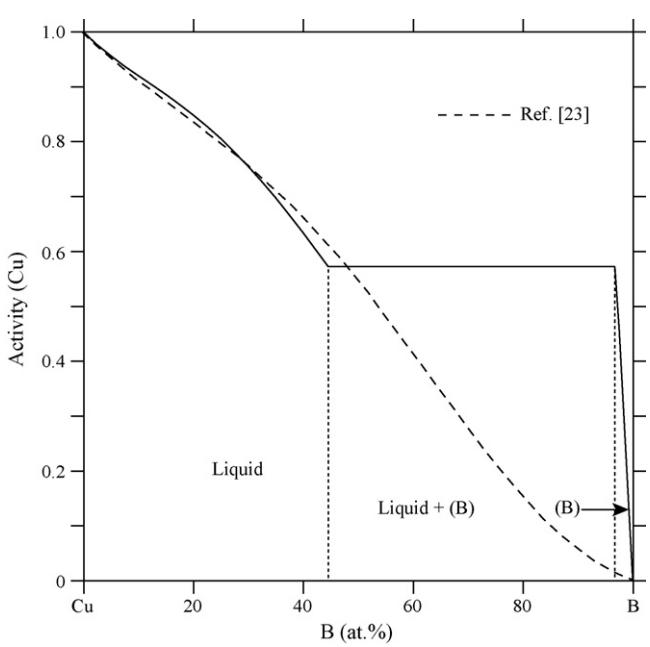

Fig. 6. The calculated activity of Cu at 1450 °C with respect to Cu (liquid) and B (β rhombohedral) compared with derived data by Jacob et al. [23].

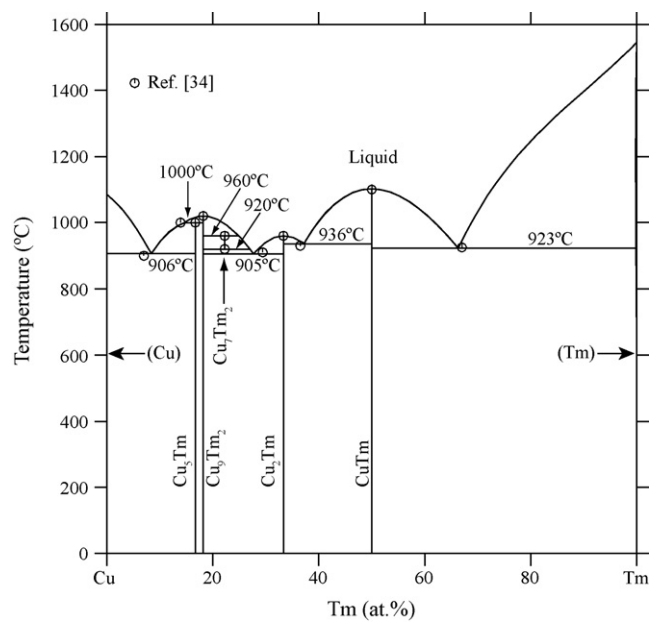

Fig. 7. The calculated Cu–Tm phase diagram with experimental data [34].

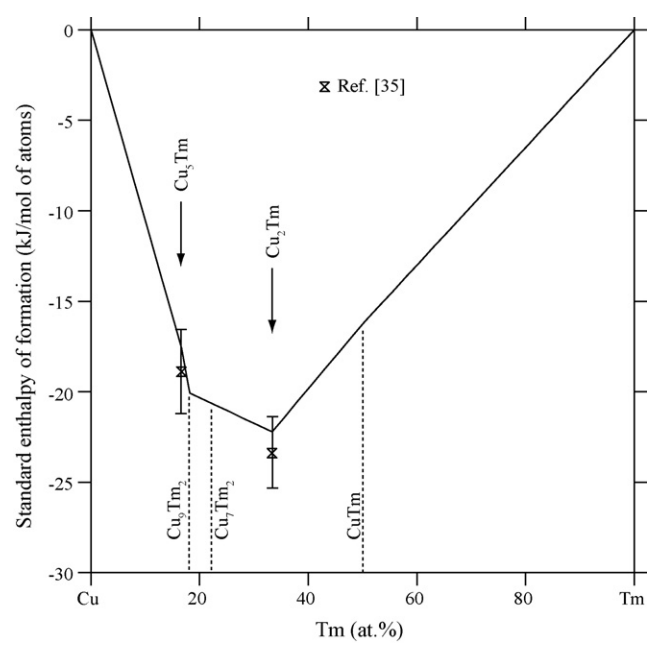

Fig. 8. The calculated standard enthalpies of formation of intermetallic compounds compared with experimental data [35].

Table 4
Invariant reactions in the Cu–Tm system.

Reaction	Reaction type	Composition (at.% Tm)			T (°C)	Ref.
$L \leftrightarrow (Cu) + Cu_5Tm$	Eutectic		0.02–0.03		850–870	[25]
		~7.0	~0	16.7	900	[34]
		8.4	0	16.7	906	This work
$L + Cu_9Tm_2 \leftrightarrow Cu_5Tm$	Peritectic	~13.9	18.2	16.7	1000	[34]
		14.5	18.2	16.7	1000	This work
$L \leftrightarrow Cu_9Tm_2$	Congruent		18.2		1020	[34]
			18.2		1020	This work
$L + Cu_9Tm_2 \leftrightarrow Cu_7Tm_2$	Peritectic		18.2	22.2	~960	[34]
		24.9	18.2	22.2	960	This work
$Cu_7Tm_2 \leftrightarrow L + Cu_9Tm_2$	Catatectic	22.2		18.2	920	[34]
		22.2	27.0	18.2	920	This work
$L \leftrightarrow Cu_9Tm_2 + Cu_2Tm$	Eutectic	~29.4	18.2	33.3	910	[34]
		27.7	18.2	33.3	905	This work
$L \leftrightarrow Cu_2Tm$	Congruent		33.3		960	[34]
			33.3		960	This work
$L \leftrightarrow Cu_2Tm + CuTm$	Eutectic	~36.5	33.3	50.0	930	[34]
		37.2	33.3	50.0	936	This work
$L \leftrightarrow CuTm$	Congruent		50.0		1100	[34]
			50.0		1100	This work
$L \leftrightarrow CuTm + (Tm)$	Eutectic	~67.0	50.0	~100	925	[34]
		66.3	50.0	100	923	This work

tively. Good agreement has been obtained in the Cu-rich region. However, different tendency between the calculated and experimental data was observed in the region from more than 45 at.% B, where there is the two-phase region at 1450 °C in the phase diagram and therefore, the calculated data of activity keeps constant. It is demonstrated that the experimental data for thermodynamic properties can be well described using the present thermodynamic parameters within experimental errors.

4.2. Cu–Tm system

Comparison of the calculated Cu–Tm phase diagram with experimental data is illustrated in Fig. 7. It is clear that the calculated phase diagram is in good agreement with the phase diagram reviewed by Subramanian and Laughlin [34]. The invariant reactions in the Cu–Tm system are listed in Table 4, where a good agreement between the calculated and experimental temperatures for these invariant reactions has been obtained.

The calculated standard enthalpies of formation of intermetallic phases are compared with experimental data in Fig. 8. As can be seen, the calculated results are reasonably consistent with experimental data measured by Meschel and Kleppa [35]. Moreover, as shown in Fig. 9, the calculated mixing enthalpy is in reasonable agreement with the experimental data at 1250 °C. The minimum values of mixing enthalpies of liquid in the Cu-RE system (such as Cu–Dy [38], Cu–Yb [39] and Cu–Er [40]) correspond to about 40 at.% RE and the present calculated results can fit this tendency.

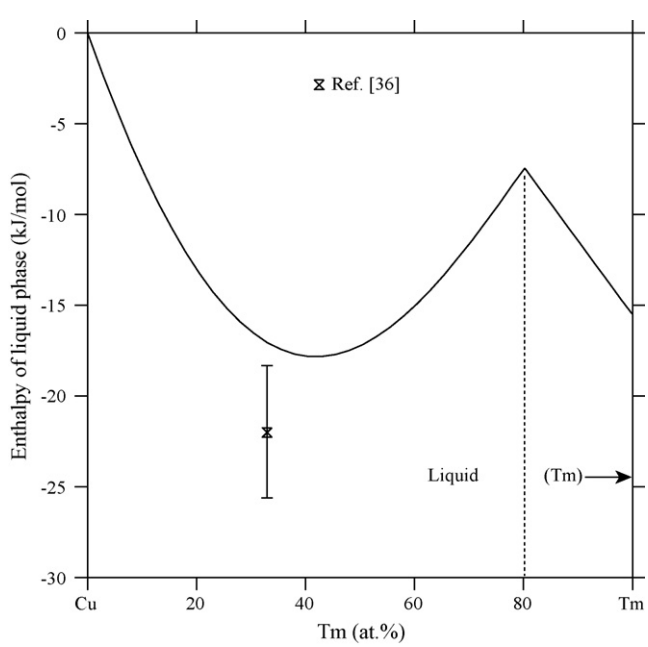

Fig. 9. The enthalpy for the alloying of liquid copper with 33 at.% Tm at 1250 °C with respect to the liquid elements [36].

5. Conclusions

The Cu–B and Cu–Tm binary systems were thermodynamically assessed based on reported experimental data of phase diagrams and thermodynamic properties. Reasonable agreement between calculated and experimental data has been reached and thermodynamic parameters for various phases in these two binary systems have been obtained.

Acknowledgements

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